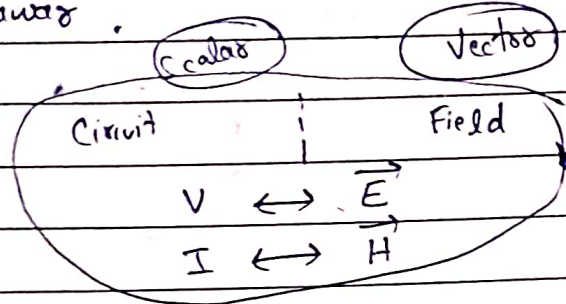


ELECTROMAGNETIC FIELD THEORY

- (1) Jordan & Palmain
- (2) SP Seth
- (3) Parul Datta



15/1/19

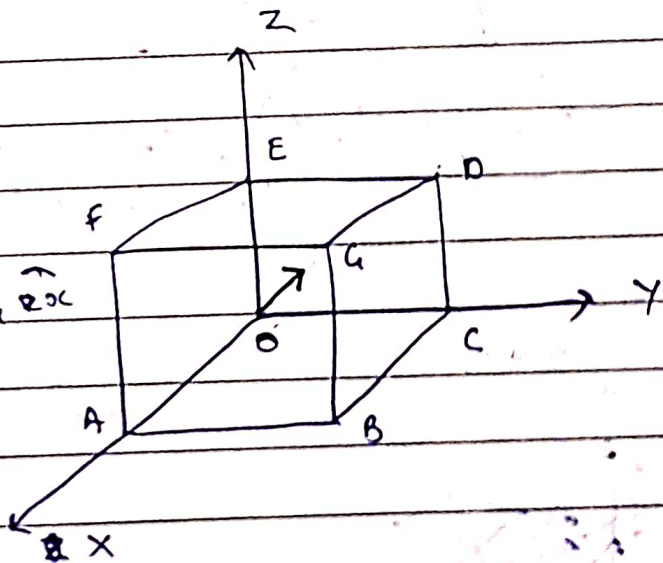
$$J = neAvd, \quad I = \int H \cdot dl$$

- incremental length
- differential length
- area dA
- volume dV

Ques Find $\vec{OQ} = ?$

Ans

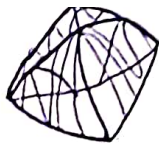
$$\vec{OQ} = \vec{OE} \hat{z} + \vec{EG} \hat{y} + \vec{GQ} \hat{x}$$



Ques

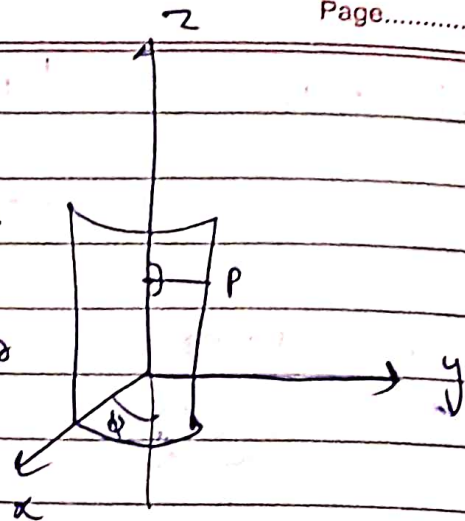
Ans

$$\vec{OQ} = 1\hat{x} + 3\hat{y} + 3\hat{z}$$



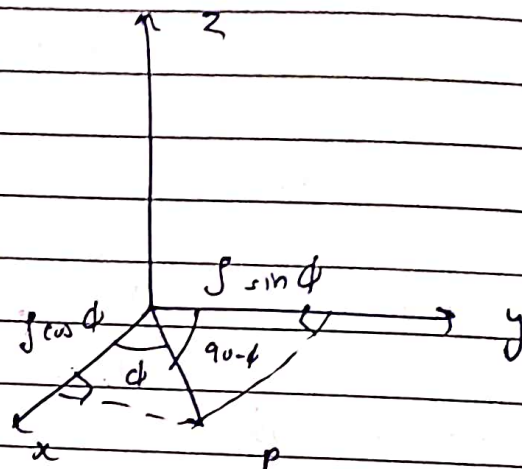
Cylindrical phase

- $\rho = \perp^*$ down from pt on z-axis
- ϕ = azimuthal angle or horizontal angle
- $z = z$



~~1. Conversion~~

$$\begin{aligned} x &= \rho \cos \phi \\ y &= \rho \sin \phi \\ z &= z \\ \phi &= \tan^{-1} \left(\frac{y}{x} \right) \\ \rho &= \sqrt{x^2 + y^2} \end{aligned}$$

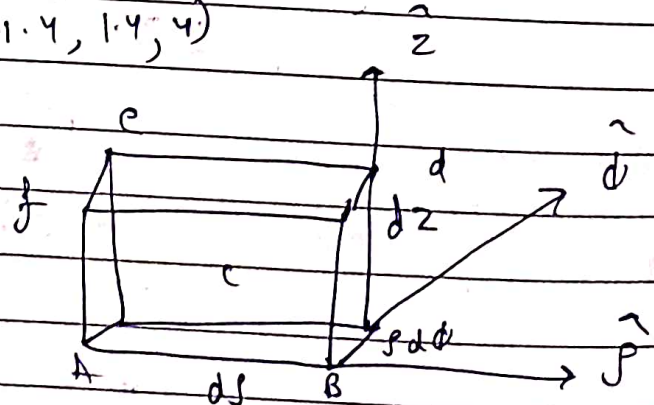
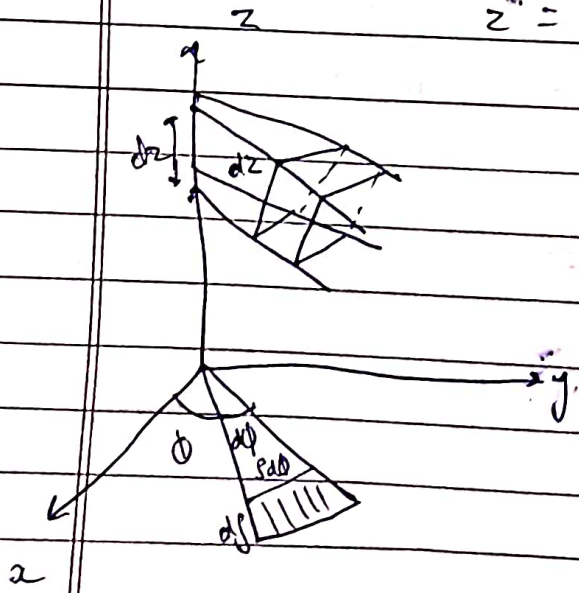


Q1:

(2, 45°, 4), Convert into Cartesian.

$$\begin{aligned} x &= 2 \cos 45^\circ = \sqrt{2} \\ y &= 2 \sin 45^\circ = \sqrt{2} \\ z &= 4 \end{aligned}$$

(1.4, 1.4, 4)



A to B

$$d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\text{Area} = dx \cdot dy \cdot \sin \theta$$

$$dV = dx \cdot dy \cdot dz$$

Ques: Point, P ($\rho=5, \phi=60^\circ, z=2$)

Q ($\rho=2, \phi=110^\circ, z=-1$)

find the dist from P to Q .. find unit vector of $\overrightarrow{PQ}$

$$P_x = 5 \times \frac{1}{2} = 2.5$$

$$P_y = 5 \times \frac{\sqrt{3}}{2} = 4.33$$

$$P_z = 2$$

$$Q_x = -0.68$$

$$Q_y = 1.87$$

$$Q_z = -1$$

$$\overrightarrow{PQ} = (-3.18, -2.46, -3)$$

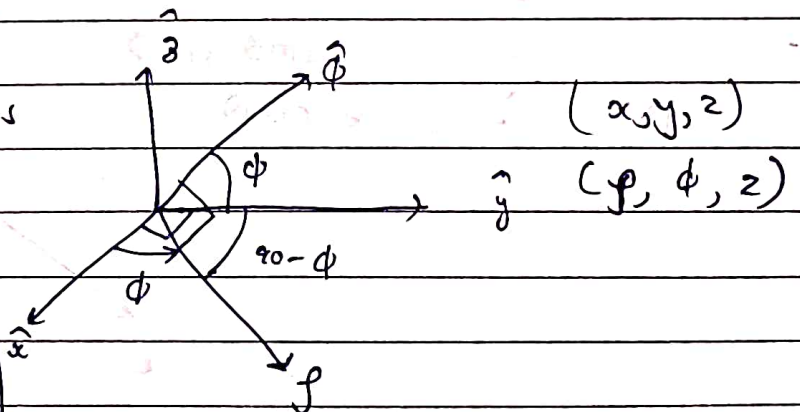
$$\text{dist} = 5$$

$$\hat{PQ} = \frac{-3.18}{5} \hat{i} - \frac{2.46}{5} \hat{j} - \frac{3}{5} \hat{k}$$

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Conversion matrix

• Cylindrical co-ordinates



Ques

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho \\ \phi \\ z \end{bmatrix}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \neq$$

Cylindrical to Cartesian

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho \\ \phi \\ z \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ \phi \\ z \end{bmatrix}$$

Cartesian
to
Cylindrical

$$= \begin{bmatrix} -2.69 \\ 3.13 \\ 7 \end{bmatrix}$$

(The ϕ will come from the
beginning vector (P))

$$\begin{bmatrix} P (2, -1, -3) \\ Q (1, 3, 4) \end{bmatrix}$$

PQ

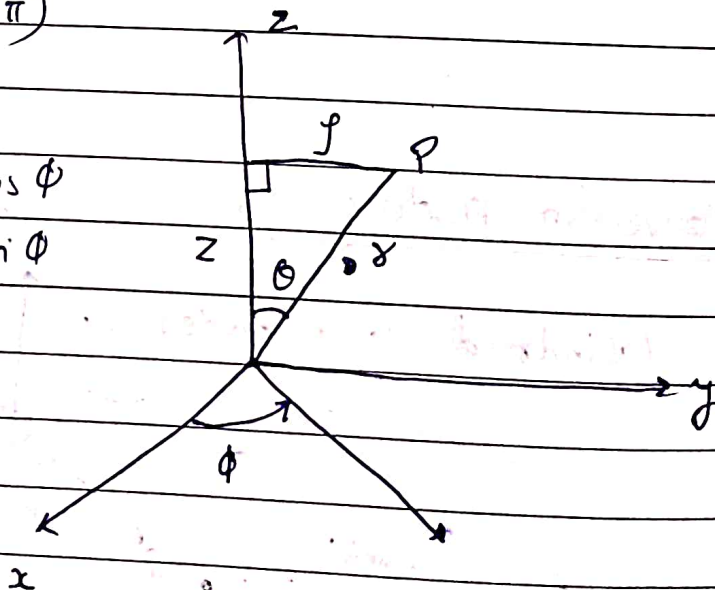
② Spherical system $\Rightarrow$

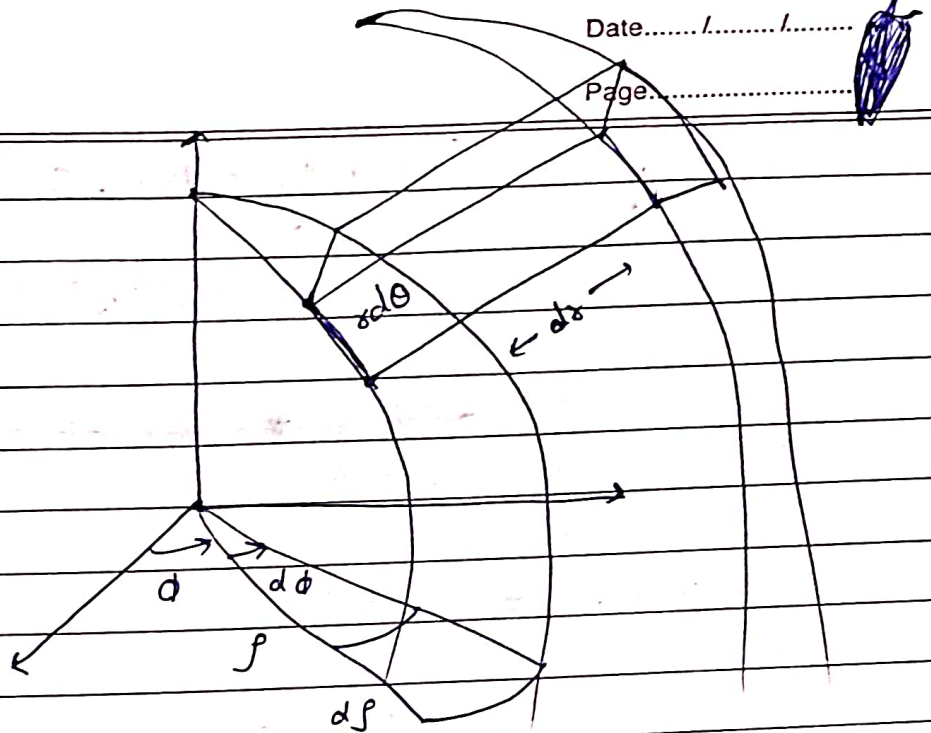
$\theta, \phi \rightarrow$ azimuth
angle
(0 to 2π)
elevation
angle
(0 to π)

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$





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$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & -\cos \phi \\ \cos \theta & \sin \theta & 0 \end{bmatrix} \begin{bmatrix} r \\ \theta \\ \phi \end{bmatrix}$$

Spherical Coordinates

$$\vec{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

Ques

Find, PQ_x , PQ_y , PQ_z .

$$P(2, -1, -3)$$

$$Q(1, 3, 4)$$

Ans

$$\theta = -26^\circ$$

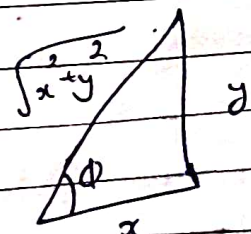
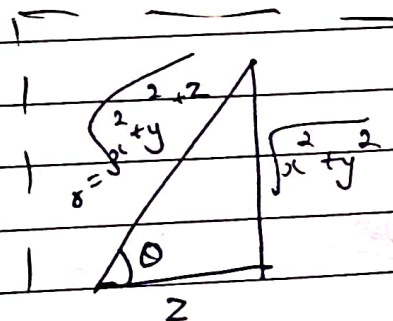
$$\phi = \cos^{-1}\left(\frac{z}{r}\right) = 143^\circ$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{14}$$

$$(PQ)_x = (-\hat{x} + 4\hat{y} + 7\hat{z}) \cdot \hat{r}$$

$$= -\sin \theta \cos \phi + 4 \sin \theta \sin \phi + 7 \cos \theta$$

$$= -7.18$$



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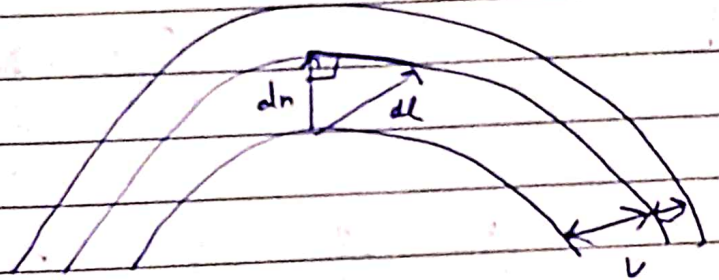
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① Gradient $\rightarrow (\nabla V)$

$$\nabla = \left(\frac{d}{dx} \hat{x} + \frac{d}{dy} \hat{y} + \frac{d}{dz} \hat{z} \right)$$

$$\cos \theta = \frac{dn}{dl}$$

$$\frac{dv}{dn} > \frac{dv}{dl}$$



Ques $V = 2x^2y - 5z$ & $P(-4, 3, 6)$, find Potential & Electric Field

Ans

$$\begin{aligned} V &= 2(16) \times 3 - 5 \times 6 \\ &= 96 - 30 \\ &= 66 \text{ V} \end{aligned}$$

$$E = \frac{dv}{dl} = (4xy - 5) \hat{x} + (2x^2) \hat{y} + (-5) \hat{z}$$

$$= 4(-4)(3) \hat{x} + 2(16) \hat{y} - 5 \hat{z}$$

$$= -48 \hat{x} + 32 \hat{y} - 5 \hat{z}$$

$$E = \sqrt{48^2 + 32^2 + 5^2} = 57.9 \text{ V}$$

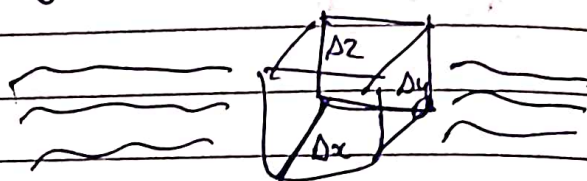
② Divergence $\rightarrow (\nabla \cdot \vec{V})$

net outward flow of liquid per unit volume

$$= \left(\frac{d}{dx} \hat{x} + \frac{d}{dy} \hat{y} + \frac{d}{dz} \hat{z} \right) \cdot (V_x \hat{x} + V_y \hat{y} + V_z \hat{z})$$

$$= \left(\frac{dV_x}{dx} + \frac{dV_y}{dy} + \frac{dV_z}{dz} \right)$$

$$\left(\frac{m}{vol} \cdot \frac{d}{t} \cdot A \right)$$



LH) $\int_m V_y \cdot \Delta x \Delta z$

$$\int_m V \Delta x \Delta z$$

$$\frac{d}{dy} \left(\int_m V_y \right) + \frac{d}{dx} \left(\int_m V_x \right) + \frac{d}{dz} \left(\int_m V_z \right)$$

Ques

A ~~vec~~ $\vec{D} = e^{-x} \sin y \hat{x} - e^{-x} \cos y \hat{y} + 2z \hat{z}$
find divergence at (origin (0,0,0))

Ans

$$\nabla \cdot \vec{D} = \left(\frac{d}{dx} \hat{x} + \frac{d}{dy} \hat{y} + \frac{d}{dz} \hat{z} \right) \cdot (e^{-x} \sin y \hat{x} - e^{-x} \cos y \hat{y} + 2z \hat{z})$$

$$= e^{-x} \cos y \sin y + e^{-x} \sin y + 2$$

$$= e^{-x} \sin y + e^{-x} \sin y + 2$$

$$= 2$$

$$\boxed{\nabla \cdot \vec{V} = 0} \rightarrow \text{Solenoidal}$$

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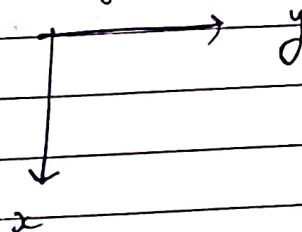
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③ Curl $\rightarrow$

rotation of a vector

rotation.



$$\boxed{\nabla \times \vec{A} = 0} \rightarrow \text{irrotational}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ u_x & u_y & u_z \end{vmatrix}$$

Q1) Find the value of c for which the vector
 $\vec{A} = (x + 3y)\hat{x} + (y - 2z)\hat{y} + (x + cz)\hat{z}$.

Ans

$$\nabla \cdot \vec{A} = 1 + 1 + c$$

$$\boxed{c = -2}$$

Q2)

$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$, Find $\nabla \cdot \vec{r}$ & $\nabla \times \vec{r}$.

Ans

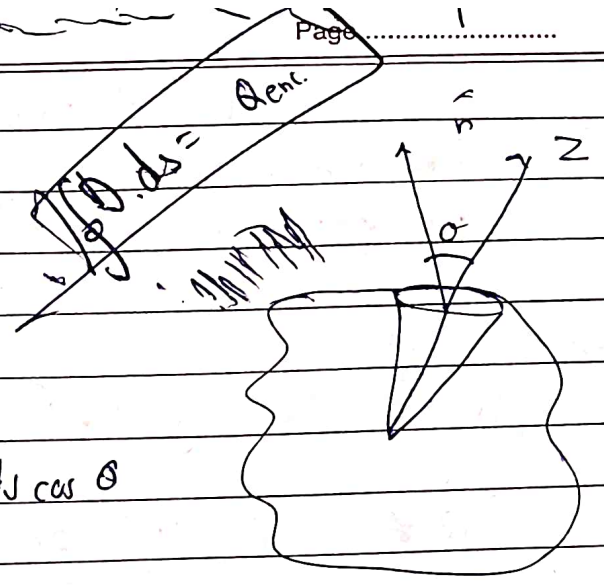
$$\nabla \cdot \vec{r} = 1 + 1 + 1 = 3$$

$$\nabla \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ x & y & z \end{vmatrix} = (0-0)\hat{x} + (0-0)\hat{y} + (0-0)\hat{z} = 0$$

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Gauss Law

flux = $\frac{q}{\epsilon_0}$ enclosed



$Q = \iiint \rho dv$

$E \cdot dS$
 $\int \frac{kq}{r^2} \cdot dS \cos \theta$
 $kq \int \frac{dS \cos \theta}{r^2}$

$\downarrow 4\pi \Rightarrow kq 4\pi = \frac{1}{4\pi\epsilon_0} \times q 4\pi = \frac{q}{\epsilon_0}$

$\iiint E \cdot dS = \frac{q}{\epsilon_0}$

$\therefore \text{flux} = \frac{q}{\epsilon_0}$

Hence, proved.

from divergence theorem, $\int \nabla \cdot E dv = \frac{q}{\epsilon_0}$

$D = E \epsilon_0$

$\int \nabla \cdot E dv = \frac{q}{\epsilon_0}$

$\int \nabla \cdot D dv = \int \rho_v dv$

$\Rightarrow \boxed{\nabla \cdot D = \rho_v}$

Imp
 Gauss divergence theorem

Ques Given the field $D = 2xy\hat{x} + x^2\hat{y}$,
 $x = 0 \text{ \& } 1$
 $y = 0 \text{ \& } 2$
 $z = 0 \text{ \& } 3$

Find $\int E \cdot dS$

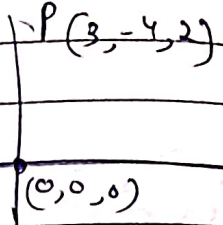
Ans
 $\int_0^2 \int_0^1 (2xy + x^2) dy dx + \int_0^2 \int_0^1 (2xy + x^2) dx dy (-1)$
 $+ \int_0^3 \int_0^2 (2xy + x^2) dy dz + \int_0^3 \int_0^2 (2xy + x^2) dy dz (-1)$

Ques Calculate the electric field at a point $(3, -4, 2)$
i) because of a charge $q_1 = 2 \text{ mC}$ at point $(0, 0, 0)$
ii) $q_2 = 3 \text{ mC}$, placed at $(-1, 2, 3)$

Ans

(i) $\vec{OP} = 3\hat{i} - 4\hat{j} + 2\hat{k}$

$OP = \frac{3\hat{i} - 4\hat{j} + 2\hat{k}}{\sqrt{16 + 9 + 4}} = \frac{3\hat{i} - 4\hat{j} + 2\hat{k}}{\sqrt{29}}$



$r = \sqrt{29}$

$\vec{E} = \frac{kq}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6}}{29} \cdot \left(\frac{3\hat{i} - 4\hat{j} + 2\hat{k}}{\sqrt{29}} \right)$

$\vec{E} = 0.6 \times 10^3 \cdot (\quad)$

$= 620 \cdot \left(\frac{3\hat{i} - 4\hat{j} + 2\hat{k}}{\sqrt{29}} \right)$

$\vec{E} = 345\hat{x} - 460\hat{y} + 2302\hat{z}$

(ii) $\vec{E} = 280\hat{x} - 419\hat{y} + 69.9\hat{k}$

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Imp

• Laplacian operator (∇^2)

$\nabla^2 V = 0 \Rightarrow \frac{d^2V}{dx^2} + \frac{d^2V}{dy^2} + \frac{d^2V}{dz^2} = 0$

$\rightarrow V = Ax + B$

$\frac{d^2V}{dx^2} = 0$
 $x \leftarrow x \leftarrow x \leftarrow x \leftarrow x = d \quad V = +V_0$
 $A = \frac{V_0}{d}$

$\therefore V = \frac{V_0 x}{d}$

$V = 0 \Rightarrow x=0 \quad V=0$
 $B = 0$

Equation of Continuity

$$I = - \frac{dq}{dt} = \frac{d\rho_v}{dt} dv$$

$$\int I \cdot ds = \int \nabla \cdot J dv$$

(By divergence theorem)

$$\therefore \boxed{\nabla \cdot J = - \frac{d\rho_v}{dt}}$$

NOTE Dielectric has more relaxation time compare to conductor.

• Relaxation time

$$\sigma (DE) = - \frac{d\rho_v}{dt}$$

(also known as scattering time)

$$\sigma \frac{\rho_v}{\epsilon} = - \frac{d\rho_v}{dt}$$

$$\int - \frac{\sigma}{\epsilon} dt = \int \frac{d\rho_v}{\rho_v}$$

$$- \frac{\sigma}{\epsilon} t + \text{const} = \ln(\rho_v)$$

$$- \frac{\sigma}{\epsilon} t + \ln(\rho_{v \text{ int}}) = \ln(\rho_v)$$

$$\ln\left(\frac{\rho_v}{\rho_{v \text{ int}}}\right) = - \frac{\sigma}{\epsilon} t$$

$$\rho_v = \rho_{v \text{ int}} e^{-\frac{\sigma}{\epsilon} t}$$

$$\Rightarrow \boxed{t_r = \frac{\epsilon}{\sigma}} \quad \left| \quad \begin{array}{c} \text{Initial charge distribution} \\ \text{Final charge distribution} \end{array} \right. \Rightarrow \begin{array}{c} \text{Initial charge distribution} \\ \text{Final charge distribution} \end{array}$$

• For good conductors relaxation time is very short. as the charge particles move very rapidly inside them and redistribute themselves at the surface quickly.

$$\rightarrow |E = \frac{t_0 \times 10}{\dots} \rightarrow$$

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- For good dielectrics the relaxation time is very long as the charge particles are bound and thus are not free to move in dielectrics.

Equation of continuity $\rightarrow$ The law of charge conservation states that the total current coming out through the surface ~~and~~ bounding a given volume must be equal to the rate of decrease of charge in volume.

(Ques) The conductivity 4.55×10^6 , $\epsilon_r = 6.9$, calculate the relaxation time assuming that the charge density is placed inside a block of sd. Determine the charge density at 2 & 3 time constants.

$$t_0 = \frac{6.9 \times 8.85 \times 10^{-12}}{4.55 \times 10^6}$$

$$t_0 = 13.4 \times 10^{-18}$$

$$f = f_0 e^{-2} = 0.135 f_0$$

$$f = f_0 e^{-3} = 0.049 f_0$$

Polarisation

$$\boxed{D = \epsilon_0 E + P} \rightarrow \text{Polarisation}$$

electric
flux density

- Bonding of charges is caused by displacement that occurs in molecules due to polarization

Ques Determine the polarization and total dipole moment in a dielectric slab which is placed under the influence of electric flux density $20\hat{x}$. The volume of the slab is 2mm . Assume the relative permittivity is 3.

Ans

$$P = D - \epsilon_0 E$$

$$= 20 - 8.85 \times 10^{-12} \times 3$$

$$P = \epsilon_0 \epsilon_r E - \epsilon_0 E = \epsilon_0 \left[1 - \frac{1}{\epsilon_r} \right] D$$

14/2/19

• Maxwell equation

1)

$$\oint \mathbf{E} \cdot d\mathbf{l} = - \frac{d\Phi}{dt} = - \frac{d(\int \mathbf{B} \cdot d\mathbf{s})}{dt}$$

↳ law of magnetic induction.

$$\oint (\nabla \times \mathbf{E}) \cdot d\mathbf{s} \Rightarrow \nabla \times \mathbf{E} = - \frac{d\mathbf{B}}{dt}$$

2)

$$\nabla \cdot \mathbf{D} = \rho_v$$

'Gauss Law'

in free space, $\nabla \cdot \mathbf{D} = 0$

then it is solenoidal

3)

Magnetic monopoles do not exist,

$$\int \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

4) Ampere's Circuital law,

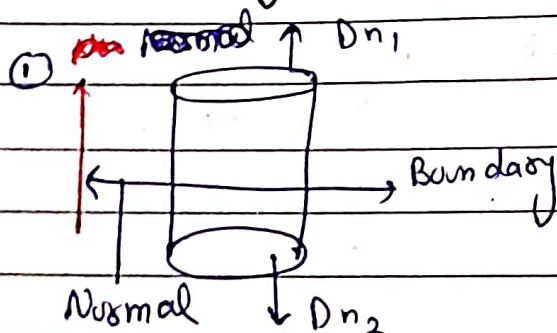
$$\oint H \cdot dl = I \rightarrow \text{(Conduction + Displacement.)}$$

$$\int I \cdot ds \quad \int \frac{dD}{dt} \cdot ds$$

$$\int (\nabla \times H) \cdot ds =$$

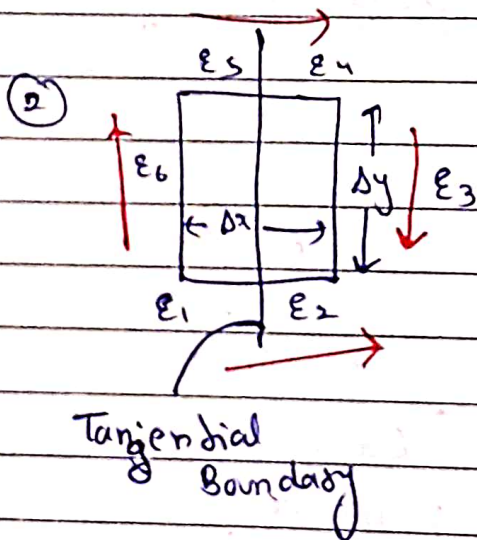
$$\nabla \times H = J + \frac{dD}{dt}$$

Now, Boundary Conditions :



$$D_{n1} da + D_{n2} da (\text{out}) + (-D_{n2}) da = \int \rho_v dv = \int \Delta x \cdot da$$

$$D_{n1} = D_{n2} \quad D_{n1} - D_{n2} = \int \Delta x$$



$$E_1 \frac{\Delta x}{2} + E_2 \frac{\Delta x}{2} + E_3 \Delta y$$

$$- E_4 \frac{\Delta x}{2} - E_5 \frac{\Delta x}{2} - E_6 \Delta y$$

$$= - \frac{dD}{dt} \cdot \Delta x \Delta y$$

$$E_{t1} = E_{t2}$$

$$E_3 = E_6$$

• Law of Refraction

first boundary condition,

$$E_{t1} = E_{t2}$$

$$\cos(90 - \theta_1) = \frac{E_{t1}}{E_1}$$

$$\boxed{E_1 \sin \theta_1 = E_2 \sin \theta_2} \quad \text{--- (1)}$$

second boundary condition,

$$D_{n1} = D_{n2}$$

$$\epsilon_{r1} E_{n1} = \epsilon_{r2} E_{n2}$$

$$\boxed{\epsilon_{r1} E_1 \cos \theta_1 = \epsilon_{r2} E_2 \cos \theta_2} \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2}$$

$$\boxed{\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}}$$

Ques $\epsilon_{r1} = 6, \epsilon_{r2} = 3, \& \quad 5\hat{x} - 2\hat{y} + 6\hat{z}$

(i) Find E_2 2, (ii) find angle which E_{t2} field make with interface, (iii) Energy density

Ans (i) $E_2 = E_{t2} + E_{n2}$
 $= 5\hat{x} - 2\hat{y} + 12\hat{z}$

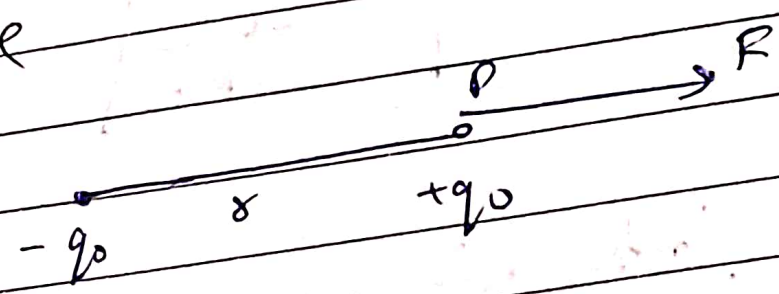
$$\begin{aligned} E_2 &= E_{t2} + E_{n2} \\ &= 5\hat{x} - 2\hat{y} + 12\hat{z} \end{aligned}$$

(ii) $\sin \alpha_1 = \frac{6}{\sqrt{25+4+36}}$

(iii) $\boxed{\text{Energy density} = \frac{1}{2} \epsilon E^2} \rightarrow \text{Energy density.}$

UNIT - 1

* Electric field due to point charge



Acc. to Coulomb law

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} \quad \text{--- (1)}$$

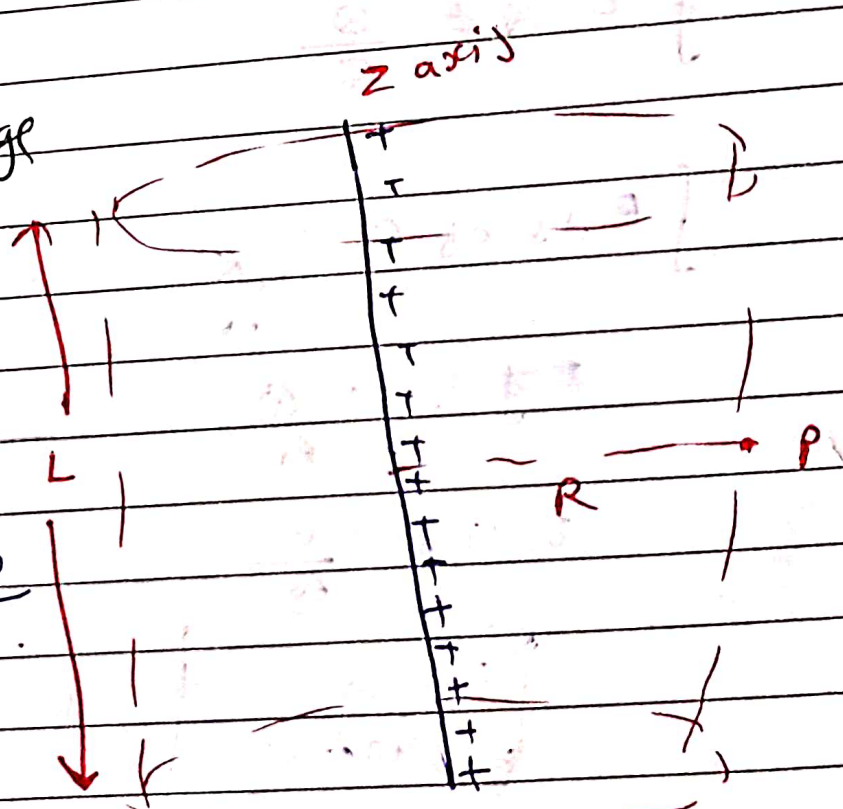
By, $E = \frac{F}{q_0}$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

* Electric field due to line charge

$$\oint \vec{E} \cdot d\vec{l} = \frac{Q}{\epsilon_0}$$

$$\left(\int_{\text{top surface}} + \int_{\text{bottom}} + \int_{\text{curved}} \right) \vec{E} \cdot d\vec{l} = \frac{Q}{\epsilon_0}$$



$$\int_{\text{top}} \vec{E} \cdot d\vec{l} = \int E dl \cos 90^\circ = 0$$

$$\int_{\text{bottom}} \vec{E} \cdot d\vec{l} = \int E dl \cos 90^\circ = 0$$

$$E \cdot 2\pi R \cancel{K} = \frac{\int_L K}{\epsilon_0}$$

$$E = \frac{\int_L}{2\pi R \epsilon_0}$$

* Electric field intensity due to conducting sphere

Case I $\rightarrow r > R$

Case II $\rightarrow r = R$

Case III $\rightarrow r < R$

Case I ($r > R$)

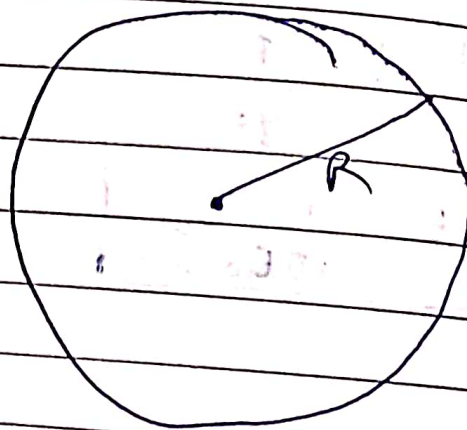
$$\oint E \cdot ds = \frac{Q}{\epsilon_0}$$

$$\oint E ds \cos 0 = \frac{Q}{\epsilon_0}$$

$$E \oint ds = \frac{Q}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi \epsilon_0 r^2}$$



Case II ($r = R$)

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$E \oint ds = \frac{Q}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

∴ ($r = R$)

$$E = \frac{Q}{4\pi \epsilon_0 R^2}$$

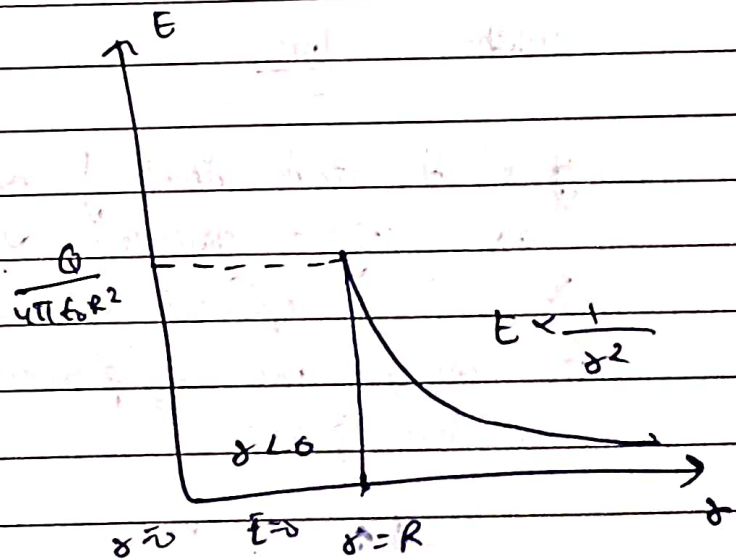
Case III ($r < R$)

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$Q_{enc} = 0$$

$$\oint \vec{E} \cdot d\vec{s} = 0$$

$$E = 0$$

★ Electric Potential

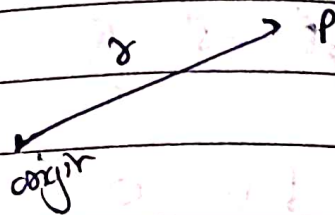
$$\int dw = w = - \int_A^B \vec{E} \cdot d\vec{l}$$

$$V_{AB} = - \int_A^B \vec{E} \cdot d\vec{l}$$

→ Potential difference.

Potential due to point charge at origin $(0,0)$

$$V = \frac{Q}{4\pi\epsilon_0 r}$$



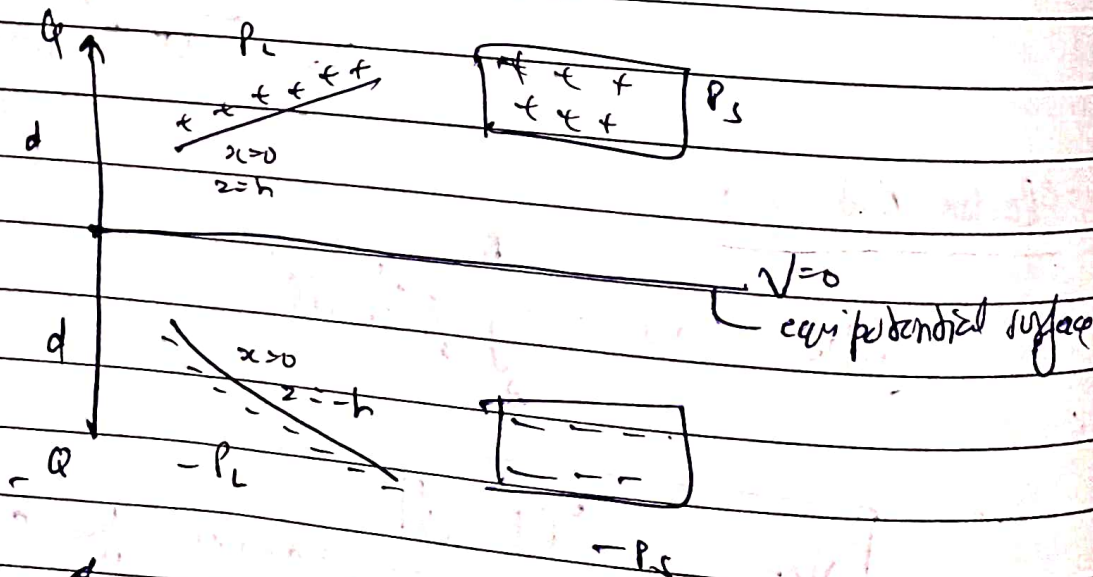
① $V_{AB} = -V_P$, work is done by field

② $V_{AB} = +V_P$, work done by external agent

③ $\frac{J}{C}$ or volts.

* Method of Images

It states that if any charge configuration placed above infinite grounded perfect conducting plane then it is replaced by charge configuration itself and its image and equipotential surface is placed in grounded plane.



* Capacitance

$$C = \frac{Q}{V_{21}}$$

① Parallel Plate Capacitor $\rightarrow$

$$Q = \int \rho_s ds$$

$$= \int \rho_s ds$$

$$Q = \int \rho_s A$$

$$V_{21} = - \int_2^1 \vec{E} \cdot d\vec{l}$$

$$\vec{E} = \frac{\int_1}{2\epsilon_0} (-\frac{Q}{A} \hat{a}_z) - \frac{\int_2}{2\epsilon_0} (Q \hat{a}_z)$$

$$= - \frac{\int_1}{\epsilon_0} \hat{a}_z$$

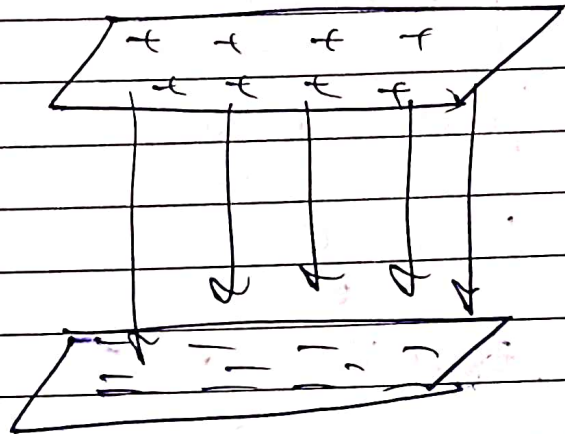
$$\vec{E} = - \frac{Q}{A\epsilon_0} \hat{a}_z$$

$$V_{21} = - \int_2^1 \frac{Q}{A\epsilon_0} \hat{a}_z \cdot d\vec{l}$$

$$V_{21} = \int_0^d \frac{Q}{A\epsilon_0} dz = \frac{Q}{A\epsilon_0} [z]_0^d$$

$$V_{21} = \frac{Qd}{A\epsilon_0}$$

$$C = \frac{Q}{V_{21}} \Rightarrow C = \frac{\epsilon_0 A}{d}$$



② Coaxial capacitor / cable

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\int E ds \cos 0 = \frac{Q}{\epsilon_0}$$

$$E \oint ds = \frac{Q}{\epsilon_0}$$

$$E \cdot 2\pi(r)L = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{2\pi\epsilon_0 r L}$$

$$V_{21} = - \int \vec{E} \cdot d\vec{l}$$

$$= - \int_a^b \frac{Q}{2\pi\epsilon_0 L} \hat{a}_r \cdot d\vec{r} \hat{a}_r$$

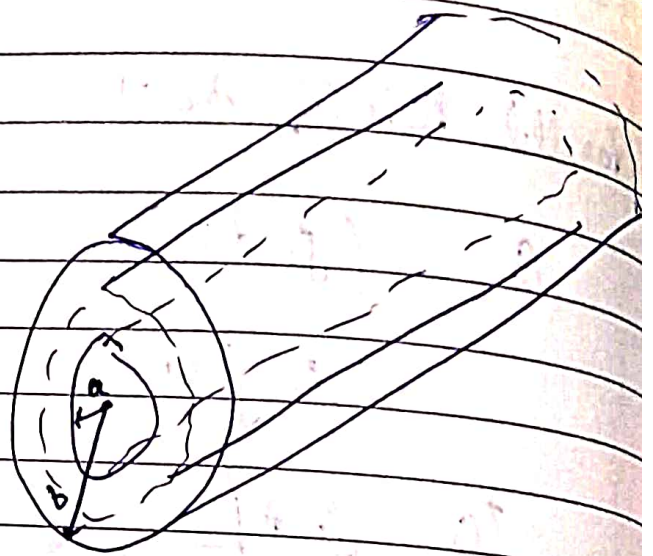
$$= - \frac{Q}{2\pi\epsilon_0 L} \ln[r]_b^a$$

$$= - \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{a}{b}\right)$$

$$C = \frac{Q}{V_{21}}$$

$$\Rightarrow C = \frac{2\pi\epsilon_0 L}{-\ln(a/b)}$$

$$C = \frac{2\pi\epsilon_0 L}{\ln(b/a)}$$



Poisson's and Laplace's equations

By Gauss's theorem,

$$\oint \mathbf{D} \cdot d\mathbf{s} = Q$$

If charge is distributed over volume,

$$\oint \mathbf{D} \cdot d\mathbf{s} = \int \rho_v dV$$

Apply Gauss divergence theorem,

$$\lim_{\Delta V \rightarrow 0} \frac{\oint \mathbf{D} \cdot d\mathbf{s}}{\Delta V} = \lim_{\Delta V \rightarrow 0} \frac{\int \rho_v dV}{\Delta V}$$

$$\boxed{\nabla \cdot \mathbf{D} = \rho_v}$$

In a charge free region, $\rho_v = 0$

$$\nabla \cdot \mathbf{D} = 0$$

It is a point form of Gauss's law,

$$\nabla \cdot \mathbf{D} = \nabla \cdot \epsilon_0 \mathbf{E} = \rho_v$$

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon_0}}$$

Electric field intensity can be expressed as negative gradient of potential,

$$E = -\nabla V$$

$$-\nabla \cdot E = \nabla \cdot \nabla V = -\frac{\rho_v}{\epsilon_0}$$

In cartesian coordinates,

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_v}{\epsilon_0}$$

This is called Poisson's equation.

For a charge free space $\rho_v = 0$

$$\boxed{\nabla^2 V = 0}$$

This is called Laplace's equation.

★ Energy stored in electric field

$$W_c = \frac{1}{2} D \cdot E \quad \text{J/m}^3$$

$$W_c = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \frac{D^2}{\epsilon}$$

$$\boxed{D = \epsilon \cdot E}$$

Unit - II

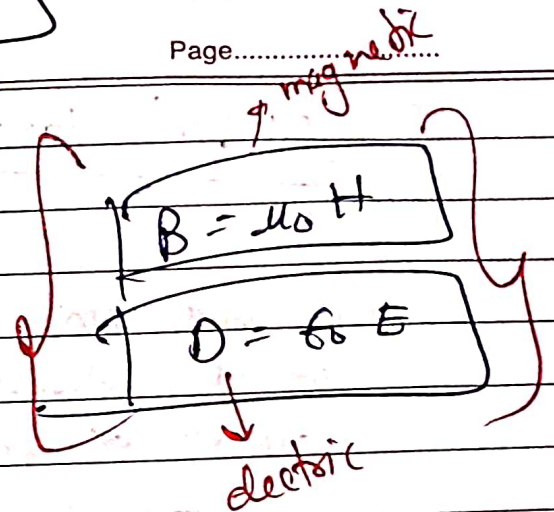
$$B = \mu H$$

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* Magnetic Induction

$$dF = qv \times B$$



* Faraday's Law

Induced emf in any closed circuit is equal to the rate of change of flux linkage of the circuit.

$$V_{emf} = \frac{d\phi}{dt}$$

$$= \frac{d(N\psi)}{dt}$$

$$= -N \frac{d(\vec{B} \cdot d\vec{S})}{dt} \quad \text{--- (1)}$$

(ϕ = Magnetic flux linkage)

$$V_{emf} = \oint \vec{E} \cdot d\vec{l} \quad \text{--- (2)}$$

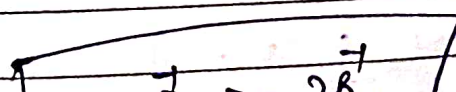
$$\oint \vec{E} \cdot d\vec{l} = -N \frac{d(\vec{B} \cdot d\vec{S})}{dt}$$

By Stokes theorem
 $\oint \vec{E} \cdot d\vec{l} = \int (\nabla \times \vec{E}) \cdot d\vec{S}$

$$\int (\nabla \times \vec{E}) \cdot d\vec{S} = -N \frac{d}{dt} \int \vec{B} \cdot d\vec{S}$$

$$\nabla \times \vec{E} = -N \frac{d\vec{B}}{dt}$$

if no. of turns $N = 1$



* Ampere's law

It states that "line integral of magnetic field intensity around closed loop is equal to the total current enclosed by that closed loop."

$$\oint \vec{H} \cdot d\vec{l} = I_{enc.}$$

$$\text{As, } I = \int \vec{J} \cdot d\vec{J}$$

$$\text{So, } \oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{J}$$

(Maxwell's 3rd equation in integral form)

By Stokes theorem,

$$\int (\nabla \times \vec{H}) \cdot d\vec{s} = \int \vec{J} \cdot d\vec{s}$$

$$\nabla \times \vec{H} = \vec{J}$$

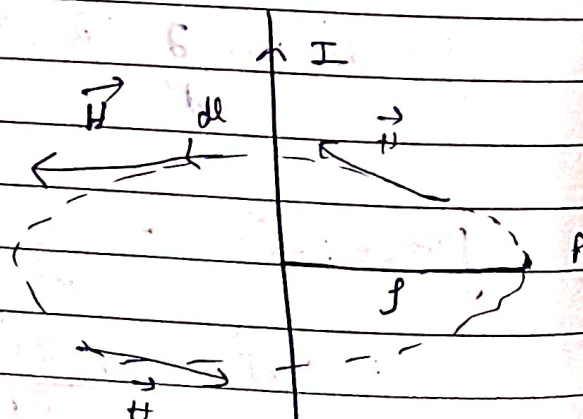
(Maxwell's 3rd eq. in differential form)

Proof:

$$\oint \vec{H} \cdot d\vec{l} = I_{enc.}$$

$$\oint H \, dl \cos 0^\circ = I$$

$$H \oint dl = I$$



$$H (2\pi r) = I$$

$$H = \frac{I}{2\pi r}$$

* Inconsistency for ampere's law

Previous Ampere's law,

$$\nabla \times H = J$$

Take divergence,

$$\nabla \cdot (\nabla \times H) = \nabla \cdot J = 0$$

$$\nabla \cdot J = -\frac{\partial \rho}{\partial t}$$

↓
Can be true only if $\frac{\partial \rho}{\partial t} = 0$,

This is unrealistic limitation,

Let us add 'G' unknown term,

$$\nabla \times H = J + G$$

Take divergence

$$\nabla \cdot (\nabla \times H) = \nabla \cdot J + \nabla \cdot G$$

$$0 = \nabla \cdot J + \nabla \cdot G$$

$$\nabla \cdot G = -\nabla \cdot J = \frac{\partial \rho}{\partial t}$$

f_v is related to $\rightarrow D$ (electric displacement)

$$D \cdot D = f_v$$

$$D \cdot Q = \frac{2}{2t} (D \cdot D) = \frac{D \cdot 2D}{2t}$$

$$Q = \frac{2D}{2t}$$

After modification,

$$D \times H = J + \frac{2D}{2t}$$

$\oint$

$$D \times H = J + J_d$$

$$(J \rightarrow \sigma E)$$

$$\oint H \cdot dl = i + i_d = \int J \cdot ds + \int \frac{2D}{2t} \cdot ds$$

Modified Ampere's law.

- The displacement current was missing in the ampere law, so by this above stated law, the inconsistency of ampere law is fulfilled.